Detailed Sprint Guide

Oct 15, 2026 · @Bansil

Each sprint below lists, per member, the stories, the exact files to create or change, and the check that says it is done. Paths match the module map in design-update.md (rev 4).

How to use this guide

Find your name in your sprint's table and work down your row. Each row gives you four things:

- **Story**: the backlog ID (US-n) and its points, or *E* for enabler work with no points.
- **Files**: `+` means create the file, `~` means change an existing one. Backend paths start at `backend/`, frontend paths at `frontend/src/`.
- **Changes**: the endpoints, tables, classes and screens to build. Names match `design-update.md` so that code review and docs agree.
- **Done when**: the check a reviewer runs before approving your PR.

Rules that hold in every sprint:

1. Read the story card in `product-backlog.md` and the screen in `interface-design.md` before you open a branch.
2. If your work needs something another member owns, look in the last section, *Hand-off contracts*. Build against a stub until the real code merges, and never block on it.
3. A new table or column always goes through Parth as an Alembic migration. Never edit a merged migration; add a new one.
4. Every new endpoint gets one `require_permission("...")` line and three API tests: success, 403 for a missing permission, and 404 for another workspace.

Repo layout and conventions

The repo stays one modular monolith: one FastAPI app, one Celery worker image, one React app. Folders that already exist from the old Sprints 0–1 are marked *(exists)*.

```
backend/
  alembic/versions/           0001, 0002 (exist) -> 0003 ... 0010
Parth
  app/
    api/                      one router file per area, thin
route owner
    core/                     auth.py (exists), permissions.py,
```

```
celery_app.py, config.py
    models/                    SQLAlchemy entities (exists, extended)
Parth
    schemas/                   Pydantic request/response models
route owner
    services/                  membership, roles, join_codes, channels,
                               issues, change_requests, reviews,
                               baselines, decision_rules
    ingestion/                 parsers (exist), extraction
Bansil
    intake/                    posts, drafts, ownership, concerns,
                               votes, watches, links, work_items
Bansil
    embeddings/  search/       vectors, candidates, hints, search        Heet
    classification/  lint/     LLM client, prompts, classifier, lint
Keval
    clarification/  consequences/  suggestions/  notifications/
Madhav
    realtime/  graph/          WebSocket gateway, event bus, deltas
Dhavni didi
    export/                    SRS export, open-items report
Bansil
  tests/                       unit/ (each owner), integration/ (Abhyudit)
frontend/src/
  app/                         shell, routing, query client, socket
Pranshu
  components/                  shared kit: Button, Badge, Drawer, Tabs
Pranshu
  api/                         typed client generated from OpenAPI
Pranshu
  features/<area>/             one folder per screen area (see owners)
eval/                          benchmark data and harness
Abhyudit
e2e/   load/                   Playwright and load scripts
Affan
docker/   .github/workflows/   Compose files and CI
Dhavni didi
```

Branches and PRs

- One branch per story: `us-30-composer`, or `e-s0-ws-spike` for enabler work. Branch from `main`, rebase before you open the PR.
- PR title `US-30: composer with EARS templates`. The body lists what changed, how to test it, and screenshots for UI work.

- One approval from the area owner (section 2 of the sprint plan); Parth also approves anything under `alembic/`. Squash-merge only.
- CI must be green: ruff, mypy, pytest, Vitest, the Playwright smoke test, and migration up and down.

Code rules

- Routes stay thin. Logic goes in `services/` or the module folder so that tests and Celery tasks can call it without HTTP.
- Publish real-time events only **after** the commit, through `app/realtime/bus.py`. Never publish from inside a transaction.
- No requirement or message text in audit rows or notification payloads: IDs and titles only (CR-04).
- Every AI output is schema-validated and labelled `AI` in the UI. User text is always rendered as text, never as HTML.
- Frontend data fetching goes through TanStack Query hooks in `features/<area>/api.ts`. Socket events invalidate or patch the query cache; components never open their own sockets.

Migrations

Migrations 0003 to 0010 land in the order below, which is not numeric order: 0010 ships in two parts (S4 and S5), and 0009 comes last in S6. Parth writes or reviews each one; the member who needs the table drafts the model and sends it to him by the sprint's day 3.

Migration	Sprint	Drafted by	Adds or changes	Needed by
`0003_severity_audit`	S0	Parth	Severity enum becomes low/medium/high; `escalated` audit action	Keval (US-08)
`0004_tenancy_roles`	S0	Parth	`projects`, `memberships` (with `role_id`, `ended_at`), `roles`, `role_permissions` with the 9 default roles seeded, `project_members`; `project_id` on documents and requirements; backfill one project per workspace	Everyone
`0005_collaboration`	S1 (codes), S2 (channels)	Madhav	`join_codes` (hash, expiry, max uses, status), `join_requests`, `channels`, `channel_members`, `messages` with a `tsvector` index	Madhav, Dhavni didi, Pranshu
`0006_lifecycle_runs`	S2–S3	Bansil, Dhavni didi	Requirement `state`, `owner_id`, `origin`; `requirement_drafts`; `analysis_runs`; `findings.project_id`, `analysis_run_id`, `is_stale`; `candidate_pairs.classification_status`; `node_positions`; `projects.graph_revision`	Heet, Keval, Abhijeet

Migration	Sprint	Drafted by	Adds or changes	Needed by
`0007_issues_change_requests`	S4	Dhavni didi, Bansil	`issues`, `issue_requirements`, `comments`, `change_requests` (`number`, `base_version`), `consequence_reports`, `review_requests`, `reviews`, `review_comments`, `change_request_issues`; `projects.baseline_revision`	Madhav, Abhijeet
`0010_trust_layer` (part 1)	S4	Bansil, Parth	`findings.source`, `raised_by`, `external_ref`, `rule_id`, `decision_owner_id`, `decision_rule_id`, `accepted_risk_reason`; `change_requests.proposal_revision`, `checks_revision`; `reviews.proposal_revision`; `consequence_reports.findings_delta`	US-56, US-35
`0008_notifications_baselines_social`	S5	Madhav, Parth	`notifications`, `baselines`, `baseline_items`, `votes`, `watches`, `requirement_links`, `suggestions`, `suggestion_feedback`	US-38, US-40, US-45, US-53
`0010_trust_layer` (part 2)	S5	Parth, Keval	`decision_rules`, `requirement_attributes` (index on project, entity, metric), feedback `action` values, `expert_rating`, `projects.use_feedback_examples`	US-57, US-58
`0009_change_log_work_items`	S6	Dhavni didi, Bansil	`graph_events` (backfilled from the audit log), `work_item_links`	US-55, US-48

The trust-layer migration is split into two files, `0010a` (S4) and `0010b` (S5), so that each sprint ships only the columns it uses. Every file must run `upgrade` and `downgrade` cleanly in CI on a database seeded with the benchmark.

Sprint 0 · Week 1 · Re-baseline (no points)

Sprint 0 ends with the new schema running under `docker compose up`, the map library chosen, and every contract in the last section agreed in writing.

Member	Story	Files	Changes	Done when
Parth	E	+ `alembic/versions/0003_severity_audit.py`, + `0004_tenancy_roles.py`, + `app/models/{project,membership,role}.py`, + `app/core/permissions.py`, ~ `app/core/auth.py`, ~ `app/api/deps.py`	Tables per the Migrations section; seed the 9 default roles and the permission catalogue (`requirements-spec.md` §1.22) as constants in `permissions.py`; `require_permission(name)` FastAPI dependency that loads the active membership and its role; move existing routes from `role ==` checks to it	Old RBAC and isolation tests pass on the new dependency; migrations go up and down

Member	Story	Files	Changes	Done when
Dhavni didi	E	+ `app/realtime/{gateway.py,bus.py,events.py}`, + `app/graph/schemas.py`, ~ `docker/compose.yml`, ~ `.github/workflows/ci.yml`	Spike: `/ws` endpoint, Redis pub/sub fan-out to two API processes; event envelope `{type, workspace_id, project_id, channel_id?, entity_id, at}`; `graph.delta` Pydantic schema; map spike with Abhijeet (Cytoscape.js vs React Flow at 500 nodes, 2,000 edges); add Redis queues to Compose	An event published by API 1 reaches a socket on API 2; spike note with fps numbers in `docs/spikes/map.md`
Bansil	E	+ `app/intake/schemas.py`, + `docs/contracts/composer.md`	Composer field schema with Pranshu: `template`, `slots{trigger, system, response, ...}`, `type`, `priority`, `owner`, `group`; draft model fields; EARS templates (ubiquitous, event, state, unwanted, optional) with Keval; plan for G3 cleanup	Schema reviewed by Pranshu and Keval
Pranshu	E	+ `frontend/src/app/{App.tsx,routes.tsx,queryClient.ts}`, + `components/{Button,Badge,Drawer,Tabs,Toast}.tsx`, + `features/auth/{LoginPage,SignupWithCodePage}.tsx`, + `styles/tokens.css`	App shell with sidebar (Home, Map, Requirements, Issues, Changes, Channels, Baselines, Insights, Members) as in the design canvas; badges carry icon plus text; login and sign-up screens (static)	Shell renders every route; Storybook or a kit page shows each component
Heet	E	+ `app/embeddings/contract.py`, + `eval/latency/hints_bench.py`	Confirm all-MiniLM-L6-v2 (384 dims); measure similar-hint latency on 500 requirements; write the `analyse_text(project, text) -> list[ClassificationResult]` signature with Keval	Hint latency under 500 ms server time, recorded
Keval	E	+ `app/classification/llm_client.py`, + `prompts/classify_v1.txt`, + `app/lint/catalogue.json`, + `app/classification/budget.py` (design stub)	`LLMClient` adapter with timeout, retries and JSON-schema validation; prompt v1 with inputs inside delimiters; lint catalogue v1 entries `{id, pattern, message, hint, severity}` (vague words, passive, missing actor, TBD, unbounded terms); `BudgetGuard` interface	Adapter unit tests with a fake model; catalogue JSON validated by a schema test
Madhav	E	+ `docs/contracts/join_codes.md`, + `app/consequences/schemas.py`	Join code format (8 characters, at least 40 bits), SHA-256 hash with pepper, expiry, max uses, approval flag, never Owner or Admin; `ImpactReport` and check-result schemas	Contracts signed off by Parth and Affan
Abhijeet	E	+ `frontend/src/features/prototypes/` (throwaway)	Clickable prototypes of join, composer, issue and change-request screens for the US-22 sessions; map library chosen with Dhavni didi	Prototype link shared; map library chosen

Member	Story	Files	Changes	Done when
Abhyudit	E	+ `eval/benchmark_v2/PLAN.md`, + `eval/benchmark_v2/schema.json`	Plan for the planted-conflict SRS, 40 post-style requirements and 10 scripted change requests with expected affected sets and owners	Plan reviewed by Keval and Heet
Affan	E	+ `e2e/playwright.config.ts`, + `e2e/smoke.spec.ts`, + `backend/tests/permissions/test_matrix.py`	Playwright in CI; permission matrix test generated from the catalogue (every permission × every default role: expect 200 or 403)	Smoke test runs in CI; matrix runs, failures expected until S1

Sprint 1 · Weeks 2–3 · Joining, roles, ingestion (34 points)

Sprint 1 is done when a PM creates a code, a developer signs up with it as Developer, uploads the planted SRS and sees its requirements with printed IDs.

Member	Story	Files	Changes	Done when
Parth	US-19 (5), US-27 (5), US-24 (3), US-01 (5), US-04 (3)	~ `app/core/auth.py`, ~ `app/api/auth.py`, + `app/services/membership.py`, + `app/services/roles.py`, + `app/api/{workspaces,projects,me}.py`, ~ `app/api/documents.py`, ~ `app/api/requirements.py`	`POST /auth/signup` requires a valid code (calls Madhav's `JoinCodeService.redeem`); `POST/GET /workspaces`, `POST/GET /workspaces/{w}/projects`, `PATCH /projects/{p}`; `GET /me/permissions?project=`; upload moves under `/projects/{p}/documents`; `GET /projects/{p}/requirements` returns printed IDs (REQ-n)	Permission matrix passes for these routes; 404 across workspaces
Madhav	US-25 (5)	+ `app/services/join_codes.py`, + `app/api/join_codes.py`, + `alembic/versions/0005_collaboration.py` (join tables only), + `app/core/celery_app.py` beat entry `expire_join_codes`	`POST/GET /workspaces/{w}/join-codes`, `DELETE /join-codes/{id}`, `POST /join`, `GET/POST /workspaces/{w}/join-requests/{id}/decision`; redeem in one transaction with `SELECT ... FOR UPDATE`; Redis rate limit 10 attempts per 10 minutes per IP and user	Expired, exhausted and revoked codes are refused; approval-required codes create a pending request
Bansil	US-02 (5)	~ `app/ingestion/extract.py`, ~ `app/ingestion/parsers/*`, + `app/ingestion/segmenter.py`	Extraction writes requirements with `project_id`, `origin=document`, `state=Proposed` (or `Approved` when imported as baseline); sentence segmentation and shall-detection; printed ID numbering per project	Planted SRS gives the expected count within 5%; unit tests per parser
Dhavni didi	US-03 (3), E	~ `app/core/celery_app.py`, + `app/realtime/bus.py` (real), ~ `app/ingestion/tasks.py`	Queues `documents`, `analysis`; processing tasks publish `analysis.progress` events after commit; reconnect re-sync by REST	Progress bar moves live in two browsers

Member	Story	Files	Changes	Done when
Pranshu	US-19, US-01, US-03, US-04 (UI)	~ `features/auth/*`, + `features/projects/{ProjectSwitcher,UploadDialog}.tsx`, + `features/requirements/{RequirementsTable,api.ts}.tsx`, + `app/socket.ts`, + `app/usePermissions.ts`	Login and sign-up with code wired; project switcher; upload with progress from socket events; requirements table (ID, text, type, state); `usePermissions()` hides actions the user lacks	E2E create code, join, upload passes
Abhijeet	US-24, US-25 (UI)	+ `features/members/{WorkspaceCreate,JoinCodesTab,JoinRequestsTab,JoinPage}.tsx`, + `features/members/api.ts`	Create workspace and first project; join-codes tab with role, expiry, max uses, approval toggle, copy button, revoke; join page that accepts a code or link	Matches the Members and Login boards on the canvas
Heet	E	+ `app/embeddings/{model.py,store.py,tasks.py}`	Embedding stage after extraction; pgvector HNSW index on `requirements.embedding`	Embeddings stored for every extracted requirement
Keval	E	+ `app/classification/{classifier.py,tasks.py}`	Classifier skeleton: takes a candidate pair, calls `LLMClient`, validates output into 5 categories plus `none`; retries with backoff	Fake-model tests pass
Abhyudit	E	+ `eval/benchmark_v2/{srs_planted.docx,labels.csv}`, + `backend/tests/integration/test_upload_extract.py`	Label every planted pair; API tests for upload and extraction	Labels reviewed; tests in CI
Affan	US-27 tests	~ `backend/tests/permissions/test_matrix.py`, + `e2e/join_upload.spec.ts`, + `backend/tests/security/test_join_codes.py`	Matrix covers S1 routes; brute-force test (rate limit hits), expiry and use-limit tests; E2E code, join, upload	All green in CI

Sprint 2 · Weeks 4–5 · Detection engine, channels, custom roles (38 points)

Sprint 2 is done when the planted SRS shows findings with confidence, two browsers chat live, and a custom role "Client reviewer" can review but not propose. If US-05 task 5a is not done by day 5, US-07, US-49 and US-22 move to Sprint 3.

Member	Story	Files	Changes	Done when
Keval	US-05 (13, classification), US-07 (5)	~ `app/classification/classifier.py`, + `prompts/{conflict,duplicate,ambiguous,incomplete,dependent}_v1.txt`, + `app/classification/explain.py`, ~ `app/models/finding.py`	5a conflicts (day 4), 5b duplicates and ambiguous (day 7), 5c incomplete and dependent (day 9); each finding stores `confidence`, `explanation`, `evidence_spans`; ambiguous and incomplete are single-requirement prompts	Category tests on the labelled pairs; JSON always validates

Member	Story	Files	Changes	Done when
Heet	US-05 (candidates)	+ `app/embeddings/candidates.py`, ~ `app/embeddings/tasks.py`	Top-k neighbours by cosine above `similarity_threshold`; write `candidate_pairs`; skip pairs already classified for the same versions	Candidate recall on labelled conflicts reported
Parth	US-26 (3), US-49 (3), E	~ `app/services/membership.py`, ~ `app/services/roles.py`, + `app/api/roles.py`, + `app/services/analysis_orchestrator.py`	`PATCH/DELETE /workspaces/{w}/members/{u}`, `POST /workspaces/{w}/leave` (last owner cannot leave); `GET /permissions`, `GET/POST /workspaces/{w}/roles`, `PATCH/DELETE /roles/{id}` (defaults immutable); `PUT /projects/{p}/members/{u}`; G1 per-pair `classification_status`; orchestration chains extract, embed, candidates, classify	Removed member's socket closes (`membership.revoked`); custom-role test passes
Madhav	US-28 (3), E	+ `app/services/channels.py`, + `app/api/channels.py`, ~ `0005_collaboration.py` (channels, messages), + `prompts/clarify_v0.txt`	`POST/GET /projects/{p}/channels`, `PATCH /channels/{c}`, `POST /channels/{c}/archive`; private channels need `channel.manage`; clarification prompt drafts	Channel CRUD tests including private-channel 404
Dhavni didi	US-29 (8), E	~ `app/realtime/gateway.py`, + `app/api/messages.py`, + `app/graph/{service.py,api.py}`	`GET/POST /channels/{c}/messages` with cursor pagination, threads (`parent_id`), mentions parsed to user IDs, edit and soft delete; socket subscribe checks membership; graph API v0 `GET /projects/{p}/graph` built from stored findings	Two browsers see messages within 1 s; subscribe to another workspace's channel is refused
Bansil	E	+ `app/intake/{posts.py,drafts.py}`, + `app/api/drafts.py`, ~ `0006_lifecycle_runs.py` (drafts)	`GET/POST/PATCH/DELETE /me/drafts`; post backend that turns a requirement post message into a `Proposed` requirement; G3 cleanup of old ingestion code	Draft autosave round-trip test
Pranshu	US-28, US-29 (UI)	+ `features/channels/{ChannelList,ChannelView,MessageList,Composer,Thread}.tsx`, + `features/channels/api.ts`	Channel sidebar, message list with virtual scroll, thread drawer, @mention autocomplete, socket events patch the query cache	Matches the Channels board; keyboard-only send works
Abhijeet	US-07, US-26, US-49 (UI), US-22	+ `features/findings/{FindingCard,FindingDrawer}.tsx`, + `features/members/{MembersTab,RolesTab,RoleEditor}.tsx`	Finding card with category badge, confidence, explanation and evidence; members list with role change and remove; role editor as a permission checklist grouped by area	Custom role created from the UI passes the matrix test
Abhyudit	US-05 accuracy, US-22	+ `eval/run_benchmark.py`, + `eval/reports/s2.md`	First precision, recall and F1 per category; run two usability sessions with prototypes	Report committed

Member	Story	Files	Changes	Done when
Affan	US-22, tests	+ `backend/tests/realtime/test_ws.py`, ~ `test_matrix.py`	WebSocket auth (token in first frame), revocation closes socket, custom-role tests	Green in CI

Sprint 3 · Weeks 6–7 · Composer, live analysis, live map (37 points) → CP1

Sprint 3 ends at Checkpoint 1: a requirement written in the composer appears on every open map with its conflict edge within 2 s of the analysis, and the feed under the map says so in writing.

Member	Story	Files	Changes	Done when
Bansil	US-30 (5), US-36 (3)	~ `app/intake/posts.py`, + `app/intake/ownership.py`, + `app/api/requirements_compose.py`	`POST /projects/{p}/requirements` and `POST /channels/{c}/requirement-posts` from composer fields; render EARS slots to text server-side; `owner_id` defaults to author; `POST /requirements/{id}/ownership-requests`, stakeholder group field	Post creates requirement, message and `analysis_run` in one transaction
Pranshu	US-30, US-36 (UI)	+ `features/composer/{ComposerPage,EarsTemplatePicker,SlotFields,PreviewPane}.tsx`, + `features/requirements/{OwnerChip,OwnershipRequestDialog}.tsx`	Composer as on the Composer board: template picker, slot fields, live preview sentence, type and priority, draft autosave every 5 s	A viewer sees no composer (permission)
Heet	US-31 (8, retrieval)	+ `app/embeddings/incremental.py`, + `app/search/similar.py`	`IncrementalAnalyzer.analyse_text()` retrieval part: embed one text, top-k within project, return candidates; `POST /projects/{p}/similar` for hints	Single requirement candidates in under 1 s
Keval	US-31 (classify, budget)	~ `app/classification/tasks.py`, ~ `app/classification/budget.py`	Classify incremental candidates; `BudgetGuard` per project monthly calls, run waits as `waiting_budget` when exceeded; result card message posted in the thread	Result card in thread within 60 s

Member	Story	Files	Changes	Done when
Dhavni didi	US-31 (queue, events), US-14 (8, deltas)	+ `app/graph/{deltas.py,narrator.py}`, ~ `app/graph/api.py`, + `app/graph/positions.py`	After each run commit: bump `projects.graph_revision`, build `graph.delta {added, changed, removed}`, batch per 250 ms, publish; `GET /projects/{p}/graph?revision=`, `PUT /projects/{p}/graph/positions`; narrator turns each delta into template sentences ("REQ-41 added by Heet; conflicts with REQ-12")	Delta lag under 2 s (NFR-32)
Abhijeet	US-14 (map UI), US-10 (UI)	+ `features/map/{MapPage,GraphCanvas,NodeDrawer,ChangesFeed,Legend,TableView}.tsx`, + `features/map/useGraphDeltas.ts`, + `features/review/ReviewActions.tsx`	Map with the chosen library: nodes by state, edges by category with icon and text, filters, persisted positions, table equivalent; Changes feed under the map; review actions on findings (confirm, dismiss with reason, escalate)	500-node map stays usable; feed text matches deltas
Parth	US-37 (3), US-10 (5)	+ `app/services/lifecycle.py`, + `app/services/review_state.py`, ~ `app/api/findings.py`	State machine Proposed → Under_Review → Approved or Rejected, Implemented, Deprecated (section 7.1 of the design); review endpoints on findings with the protected-requirement guard	Illegal transitions return 409
Madhav	US-09 (5)	+ `app/clarification/{generator.py,tasks.py}`, + `app/api/clarifications.py`	Clarification questions for ambiguous and incomplete findings; posted into the requirement thread and mention the owner	Questions created for every ambiguous finding in the benchmark
Abhyudit	CP1	+ `eval/reports/cp1.md`, ~ `eval/run_benchmark.py`	Accuracy report per category, document-style and post-style separately	Report ready day 9
Affan	CP1	+ `e2e/cp1_collab.spec.ts`, + `load/map_freshness.js`	E2E: join, compose, result card, map update in a second browser; measure delta lag; tenant isolation over REST and socket	CP1 script runs end to end

Sprint 4 · Weeks 8–9 · Issues and PR-style change requests (37 points)

Sprint 4 is done when a two-owner conflict opens an issue, a change request that says *Resolves ISS-n* merges as version 2, closes the issue and updates the map, and a second open request on the same requirement shows "needs update". Week 1 builds US-11, US-33, US-52 and US-56; week 2 builds US-34 and US-35 on top of them.

Member	Story	Files	Changes	Done when
Dhavni didi	US-52 (5)	+ `app/services/issues.py`, + `app/api/issues.py`, + `alembic/versions/0007_issues_change_requests.py` (issue tables)	`GET/POST /projects/{p}/issues`, `GET/PATCH /issues/{id}` (labels, assignees, close, reopen), comments; auto conflict issue when a Conflicting finding spans two owners (one per finding, unique index); reopen when re-flagged; `issue.*` events	Auto issue assigned to both owners
Bansil	US-33 (5), US-56 (3)	+ `app/intake/change_intake.py`, + `app/intake/concerns.py`, + `app/api/concerns.py`, + `0010a_trust_findings.py`	`POST /requirements/{id}/change-requests` storing `base_version`, `proposal_revision=1`, `Resolves ISS-n` links; `POST /projects/{p}/concerns` creates a finding with `source=person`, `raised_by`, `external_ref`, opens an issue, records `ai_miss` when no AI or rule finding exists	Concern shows on the map with a Person badge
Madhav	US-11 (8), US-34 (8)	+ `app/consequences/{impact.py,checks.py,delta.py,risk.py,tasks.py}`	`ImpactAnalyzer.analyse()`: affected requirements, owners, work items with reasons; checks (new conflicts, duplicates, dependency breaks, protected, stale) each `pending/pass/warning/fail/info`; findings delta `resolved/new/still_open/must_rerun`; risk low/medium/high; report marked stale when base moves	Scripted change requests give the expected affected sets
Heet	US-11 (relations)	+ `app/search/relations.py`	One-hop and two-hop related requirements from findings, human links and shared embeddings	Used by impact; unit tests on a fixture graph
Keval	US-34 (narrative)	+ `prompts/consequence_narrative_v1.txt`, + `app/classification/narrative.py`	AI narrative from the structured report only, labelled AI	Narrative never contradicts check results (test)
Parth	US-35 (5), US-12 (3), 0010a	+ `app/services/{change_requests,reviews}.py`, + `app/services/agreement.py`, + `app/api/change_requests.py`, ~ `app/models/requirement.py`	Reviews (approve, request changes, comment, inline); auto-request reviewers; `PATCH` bumps `proposal_revision` and dismisses older approvals; `AgreementGuard`; `POST /change-requests/{id}/merge` in one transaction returning `409 {unmet}`; version history table and `GET /requirements/{id}/versions`	No self-approval; competing request goes `needs_update`

Member	Story	Files	Changes	Done when
Abhijeet	US-33, US-34, US-35 (UI)	+ `features/changes/{ChangeEditor,ChangeRequestPage,DiffView,ChecksTab,ReviewsTab,MergeBox}.tsx`	Change editor with word diff; request page with Conversation, Checks and Files-changed tabs; merge box lists unmet conditions from the 409	Starts day 1 against `fixtures/rcr_sample.json`
Pranshu	US-52, US-12, US-56 (UI)	+ `features/issues/{IssueList,IssuePage,RaiseConcernDialog}.tsx` , + `features/requirements/VersionHistory.tsx`	Issues list and page as on the Issues board; source badges AI, Rule, Person; Raise a concern dialog; version history with diff	Matches the Issues board
Abhyudit	tests	+ `backend/tests/integration/test_impact.py` , + `test_auto_issues.py`	Affected sets and findings delta for all 10 scripted requests; auto-issue owners	Green
Affan	tests	+ `backend/tests/security/test_merge_rules.py` , + `e2e/change_request.spec.ts`	Self-approval, merge permission, needs update, compliance sign-off, edit dismisses approvals (FR-98)	Green

Sprint 5 · Weeks 10–11 · Suggestions, decision rules, cross-checks (38 points) → CP2

Sprint 5 ends at Checkpoint 2: an auto conflict issue gets three AI suggestions, both owners mark a preference, one is edited and accepted into a pre-filled change request, it merges, and snapshot v1.1 is compared with v1.0. The recall report must count conflicts caught by AI, by rule checks, and missed.

Member	Story	Files	Changes	Done when
Madhav	US-45 (5), US-38 (5)	+ `app/suggestions/{service.py,schemas.py,tasks.py}` , + `app/api/suggestions.py` , + `app/notifications/{service.py,api.py}` , + `0008_notifications_baselines_social.py` (suggestions, notifications)	`GET /findings/{id}/suggestions` , `POST .../regenerate` , `POST /suggestions/{id}/feedback` (append-only, six outcomes plus prefer and rate), `GET /projects/{p}/suggestions/stats` ; suggestions never edit a requirement; `NotificationService.notify()` with grouped inbox, `GET /notifications` , `POST /notifications/read`	Feedback rows cannot be updated or deleted (DB test)

Member	Story	Files	Changes	Done when
Keval	US-45 (prompts), US-51 (lint, pre-flight, AI wording), US-58 (extraction), US-06	+ `prompts/suggest_v1.txt`, + `app/lint/{engine.py,api.py}`, + `app/classification/{preflight.py,wording.py}`, + `app/classification/attributes.py` (`AttributeExtractor`), ~ `app/api/projects.py`	Suggestion prompt returns `{options:[{text, trade_off, confidence, tbd_slots}]}`; `GET /lint/catalogue`, `POST /projects/{p}/lint`, `/preflight`, `/improve-wording` (shown as diff only); attribute extraction patterns first, LLM fallback with `extractor=llm`; per-project thresholds editable	Pre-flight limited by `BudgetGuard`
Heet	US-58 (attribute index, checks), US-06	+ `app/search/rule_checker.py` (`RuleChecker`), ~ `0010b_trust_rules.py` (`requirement_attributes`)	Numeric-limit, negation, actor and condition-overlap checks for requirements sharing entity or metric; findings with `source=rule`, `rule_id`, rationale from attributes; threshold tuning	Rule checks catch the below-threshold benchmark subset
Parth	US-57 (3), US-13 (aggregation), US-40 (5)	+ `app/services/decision_rules.py` (`DecisionRuleEngine`), + `app/api/decision_rules.py`, + `app/services/baselines.py`, + `app/api/{baselines,analytics}.py`	`GET/POST /projects/{p}/decision-rules`, `PATCH/DELETE /decision-rules/{id}`, `GET /findings/{id}/decision`, `GET /change-requests/{id}/decision`; `AgreementGuard` reads required reviewers from the rule; `POST/GET /projects/{p}/baselines`, `GET /baselines/{a}/compare/{b}`; dashboard counts query	Dashboard counts equal DB counts
Bansil	US-51 (wiring)	~ `app/intake/posts.py`, ~ `app/intake/change_intake.py`	Composer calls lint, similar and pre-flight; change editor pre-fills from an accepted suggestion with `source_suggestion_id`	Accepted suggestion opens a pre-filled request
Abhijeet	US-45 (panel), US-38, US-06 (UI)	+ `features/findings/SuggestionPanel.tsx`, + `features/notifications/{Inbox,Bell}.tsx`, + `features/members/ProjectSettings.tsx`	Three option cards with trade-off and confidence; Accept, Edit and accept, Reject with reason chips, Defer, Prefer; inbox grouped by kind; threshold sliders	Matches section 8a of the interface design
Pranshu	US-13, US-40, US-57 (UI), US-51 (UI)	+ `features/dashboard/DashboardPage.tsx`, + `features/baselines/{BaselineList,CompareView}.tsx`, + `features/members/DecisionRulesTab.tsx`, ~ `features/composer/*` (lint underline, hints, pre-flight)	Dashboard cards; baseline compare with added, changed, removed; decision-rules table; decision owner shown on every issue and request	Matches the Baselines and Members boards
Dhavni didi	load test	+ `load/{messages,analysis,checks,map}.js`	With Affan: NFR-24 to 26, 32, 33	Results in `load/reports/cp2.md`

Member	Story	Files	Changes	Done when
Abhyudit	CP2	+ `eval/benchmark_v2/below_threshold.csv`, + `eval/reports/cp2.md`, + `eval/suggestion_rating.py`	Recall by layer (AI, rules, missed); analysts rate 30 suggestions, stored as `expert_rating`	Report ready day 9
Affan	CP2	+ `e2e/cp2_full_loop.spec.ts`	Full loop E2E; load tests with Dhavni didi	CP2 script runs end to end

Sprint 6 · Weeks 12–13 · Written changes, exports, governance (35 points)

Sprint 6 ends with feature freeze at the end of Week 13; anything unfinished goes back to the backlog, not into Sprint 7.

Member	Story	Files	Changes	Done when
Dhavni didi	US-50 (5), US-42 (3), US-55 (3)	+ `alembic/versions/0009_change_log_work_items.py` (`graph_events`), ~ `app/graph/narrator.py`, + `app/graph/{preview.py,revisions.py,changes_api.py}`, + `app/api/timeline.py`	Persist every narrator sentence as a `graph_events` row (backfill from audit); `GET /projects/{p}/changes?since=&actor=&kind=&requirement=`, `GET /projects/{p}/graph/revisions`, `GET /change-requests/{id}/graph-preview`; `GET /requirements/{id}/timeline`	Every delta has at least one sentence
Keval	US-55 (AI summary), US-08 (3), US-15 (3)	+ `prompts/change_summary_v1.txt`, + `app/classification/{severity.py,feedback_loop.py}`	`POST /projects/{p}/changes/summary` with optional AI narrative built only from `graph_events`; severity low, medium, high per finding; feedback loop reports dismissed-finding reasons per category (no self-training)	Summary cites only listed events (test)
Pranshu	US-50, US-42, US-55 (UI)	+ `features/map/{PreviewLayer,TimeSlider}.tsx` (with Abhijeet), + `features/requirements/Timeline.tsx`, + `features/analytics/ChangesPanel.tsx`	Proposed layer toggle on the map, revision slider, written Changes panel with filters and "Summarise since"	Matches the Map board

Member	Story	Files	Changes	Done when
Bansil	US-41 (5), US-59 (2), US-48 (3)	+ `app/export/{srs_docx.py,srs_pdf.py,open_items.py}`, + `app/intake/work_items.py`, ~ `0009_change_log_work_items.py` (`work_item_links`)	`GET /baselines/{id}/export?format=docx	pdf `with Approved and Implemented only plus an exclusion note`; GET /baselines/{id}/open-items?format=pdf
Parth	US-18 (5, conditional)	~ `app/services/reviews.py`, ~ `app/services/decision_rules.py`, ~ `app/models/requirement.py`	`protected` flag; compliance reviewer approval required on protected requirements; default decision rule for protected	Merge without compliance approval returns 409
Madhav	US-53 (3), E2E	+ `app/intake/social.py` or `app/services/social.py`, + `app/api/social.py`	`GET/POST /{subject}/{id}/comments`, `PUT/DELETE .../vote`, `PUT/DELETE .../watch`, `POST/DELETE /requirements/{id}/links`; watchers get notifications; votes never decide (display only)	One vote per member per subject
Abhijeet	US-18, US-15, US-08 (UI)	+ `features/compliance/{ProtectedBadge,ComplianceQueue}.tsx`, ~ `features/findings/FindingCard.tsx`	Protected badge and compliance queue; severity badge; feedback reasons view	Keyboard and screen-reader pass on these screens
Heet	E	~ `app/search/rule_checker.py`, ~ `app/embeddings/candidates.py`	Recall tuning for US-58 using CP2 misses	Recall by layer improves or is documented
Abhyudit	regression	~ `eval/run_benchmark.py`, + `backend/tests/integration/test_change_log.py`	Full regression; final accuracy; change-log sentences match deltas	Final numbers in `eval/reports/final.md`
Affan	security	+ `backend/tests/security/{test_isolation,test_injection,test_rate_limits}.py`, + `e2e/a11y.spec.ts`	With Madhav: isolation, join codes, rate limits, prompt injection in posts; axe accessibility checks on core flows; production smoke	All green on staging

Sprint 7 · Week 14 · Release (no points)

Sprint 7 adds no features: only blocking fixes, demo data and the final report.

Day	Activity	Lead	Files
1	Bug bash on the deployed system; every bug filed with steps and severity	Affan, all	GitHub issues labelled `bug-bash`

Day	Activity	Lead	Files
2–3	Blocking fixes only; deployment freeze at end of day 3	Parth, Dhavni didi	Fix branches `fix-<issue>`; ~ `docker/compose.prod.yml`
3	Demo data: 1 workspace, 3 projects, 10 members in different roles, planted scenarios	Abhyudit, Bansil	+ `scripts/seed_demo.py`, + `eval/demo/*.json`
4	Demo script and two rehearsals	Pranshu, Abhijeet	+ `docs/demo-script.md`
4–5	Final report: accuracy, load, usability, retrospective, extensions	Madhav, Heet, Keval	+ `docs/final-report.md`
5	Final demo	All	—

Hand-off contracts

These are the points where one member's code waits on another's. The consumer builds against a stub or fixture until the producer merges.

Contract	Producer	Consumers	Needed by	Stub to use meanwhile
Permission catalogue and `require_permission()`	Parth	Everyone writing routes; Affan	S0 end	Constant list in `app/core/permissions.py`
`JoinCodeService.redeem(code, user)`	Madhav	Parth (sign-up)	S1 day 5	Function that accepts code `TEST1234`
Event envelope and `bus.publish()`	Dhavni didi	Pranshu, Abhijeet, Madhav	S1 day 3	Local in-process bus
`GET /me/permissions`	Parth	Pranshu, Abhijeet	S1 day 5	Hard-coded permission set per role
`IncrementalAnalyzer.analyse_text()`	Heet, Keval	Madhav (impact), Bansil	S3 day 5	Returns fixture findings
`graph.delta` schema and `GET /graph?revision=`	Dhavni didi	Abhijeet, Pranshu	S3 day 3	`fixtures/graph_delta.json`
Lint catalogue JSON	Keval	Pranshu, Bansil	S3 day 1	Catalogue v1 from S0
Issue API	Dhavni didi	Pranshu, Madhav	S4 day 4	`fixtures/issue_sample.json`
Change-request and review API	Parth	Abhijeet	S4 day 5	`fixtures/rcr_sample.json`
`ImpactAnalyzer.analyse()` and findings delta	Madhav	Parth (merge), Abhijeet (Checks tab)	S4 day 6	Fixed report JSON

Contract	Producer	Consumers	Needed by	Stub to use meanwhile
`NotificationService.notify()`	Madhav	Parth, Heet, Keval, Dhavni didi	S5 day 3	Logs instead of storing
Suggestion schema and feedback API	Madhav	Abhijeet, Keval, Bansil	S5 day 3	`fixtures/suggestions.json`
`DecisionRuleEngine.resolve()`	Parth	Bansil (concerns), Dhavni didi (issues), Pranshu	S5 day 4	Returns the project's PM as owner
`graph_events` sentence templates	Dhavni didi	Pranshu, Keval, Abhyudit	S6 day 2	Narrator output from S3

If a contract is late by more than two days, the consumer raises it at stand-up and the Scrum Master applies the cut order in `epics.md`.